A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 09

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 Choosing by eye is biased:** people are drawn to patches that look interesting, flowery or easy to reach, so the sample does not represent the whole field **1** Random numbers used as coordinates on a grid give every point an equal chance of being sampled, so the sample is representative and the mean can be applied to the whole area **1** An unbiased sample makes the comparison between the two fields valid, because any difference found is due to the grazing and not to how the sites were chosen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area • **random** → positions chosen using random numbers so the choice is not biased • **valid** → actually measures what it is meant to measure **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = (number in first sample x number in second sample) / number marked in second sample = (60 x 75) / 15 **1** = 300 grasshoppers **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Lincoln index** → estimating population size from the proportion of marked animals recaptured **Do not credit** • A bare wrong answer with no working.	2
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The marked individuals mix randomly back into the population before the second sample is taken **1** The mark does not affect the animal's survival or behaviour, is not lost, and does not make it more visible to predators or to the collector **1 The population is closed:** there are no births, deaths, immigration or emigration between the two samples **1** Marked and unmarked individuals are equally likely to be caught in the second sample **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **valid** → actually measures what it is meant to measure **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	3

Do not write outside the box

Question	Marking guidance	Marks
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** The quadrat must be large enough to contain a workable number of individuals of the species present but small enough to count accurately - a small quadrat for low-growing grassland plants, a larger one for shrubs **1 A pilot study is used:** the cumulative number of species found is plotted against the number of quadrats, and enough quadrats are taken for the curve to level off **1** More quadrats reduce the effect of chance on the mean and make the result more reliable, but the time available limits how many can be done, so a compromise is reached **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
01.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Survey both fields on the same day, or in the same weather, so that differences in temperature, rainfall or season do not affect the result **1** Use the same method, the same quadrat size and the same number of samples in each field, so the only difference between them is the grazing **1** Choose fields that match in soil type, aspect, slope and altitude, so grazing really is the only variable being tested **1** Identify the plants with the same key, and have the same person make the estimates, so judgements such as percentage cover are consistent **1** Repeat the whole survey and calculate means, and record the spread of the data so the variability within each field can be compared with the difference between them **1** Avoid trampled areas near gates and the field edges, where conditions are untypical of the field as a whole **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **reliable** → repeatable / gives consistent results when repeated • **valid** → actually measures what it is meant to measure • **quadrat** → a square frame of known area used to sample a fixed area **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-402; AQA spec 3.7.2.1-3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
02.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well – pump-jacks ('nodding donkeys') increase flow **1** Pressure falls as oil is removed so the remaining oil no longer flows to the well **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) **1** Captured CO2 can be pumped down instead, increasing oil recovery while storing the CO2 underground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground	2
03.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity **1** Controlled underground combustion to heat the oil **1** Detergents / solvents to reduce surface tension so oil flows more easily **1** Bacteria that partially digest heavy oil into lighter oils and produce CO2 to maintain pressure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** Many wells can be drilled from a single platform, reducing land take / number of rigs **1** Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached **1** Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s **1** Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (225 - 95) / 225 x 100 **1** = 57.8% (accept 58%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form **1** Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl2) during the dark winter **1** The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated **1** When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone – levels recover by December as the vortex breaks down **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation **1** Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **correlation** → a relationship in which the two variables change together • **causation** → one variable actually causing the change in the other **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	
05.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1
05.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Bacteria in the gut convert nitrates to nitrites **1** Nitrites react with haemoglobin so it cannot carry oxygen **1** Infants are most at risk because they have more of the gut bacteria and their haemoglobin reacts faster – usually from shallow wells receiving arable drainage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrites may be converted in the gut to nitrosamines which are carcinogenic **1** But most research has been on rats / the evidence in humans is not conclusive – nitrites are also ingested in processed food so the source cannot be isolated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Reduce use of nitrate fertilisers / match application to crop need **1** Avoid ploughing / applying fertiliser at times of heavy rain (autumn / winter) **1** Use low-solubility / slow-release fertilisers rather than soluble rapid-release ones **1** Store manure / slurry where it cannot leach or wash into rivers **1** Grow crops with lower nitrogen requirements (eg triticale instead of wheat) **1** Grow legumes with nitrogen-fixing bacteria instead of relying on artificial fertiliser **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **nitrogen-fixing** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Iron(III) sulfate is added to effluent: phosphates are precipitated as insoluble iron phosphate (a fine sediment that can be used as fertiliser) **1 Lakes:** dredging phosphate-rich sediments / adding iron(III) sulfate to reduce phosphate solubility (eg the Norfolk Broads) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 267-268; AQA spec 3.4.3.2.9.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Organic (methyl) mercury **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury **1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are less soluble at high pH **1** So they remain immobile and are not leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Most solar energy is absorbed at the surface in the tropics where insolation is most intense / the Sun is overhead **1** The warm surface heats the air above – warm winds (eg south-westerlies) carry heat energy to higher latitudes **1** Winds blowing over the oceans create currents (North Atlantic Conveyor / Gulf Stream) carrying warm water from the tropics to NW Europe **1** Evaporation in the tropics stores latent heat in water vapour which is released when it condenses at higher latitudes **OR** transports water to areas that would be dry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun • **latent heat** → the energy absorbed or released when water changes state	4

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** The North Atlantic Conveyor brings warm water from the tropics to the coast of NW Europe **1** Prevailing south-westerly winds blowing over the warm sea carry the heat onto the land **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Atmospheric pressure controls how easily water molecules evaporate / escape from the surface **1** At low pressure water would boil / evaporate at low temperatures so all surface water would become vapour **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Nitrogen **1** Oxygen **1** Carbon dioxide **1** Argon / neon / krypton / xenon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87, 96-97; AQA spec 3.2.1.1, 3.2.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2 Habitat change:** flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) **up to 2 Wildlife barrier:** the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river **up to 2 Changed river regime downstream:** regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds **up to 2 Sedimentation:** sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion **up to 2 Microclimate:** water's high heat capacity reduces temperature fluctuations – higher wind speeds – more evaporation, humidity, cloud and precipitation downwind **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vegetation / dead organic matter under water decays anaerobically **1** Releasing methane, a powerful greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rain surplus water is stored, reducing peak flow / flooding downstream **1** In dry periods the dam releases water to maintain river levels / supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** No reservoir range = 3.9 - 0.7 = 3.2 m – with reservoir = 2.7 - 1.7 = 1.0 m **1** Reduction = 2.2 m **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream **1** During dry periods the dam is opened to release water and maintain river levels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Deforestation / urbanisation reduce infiltration and interception **1** So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Site factors:** topography - narrow exit from a large deep basin so a small dam holds a large volume – impermeable, strong bedrock without faults or seismic activity – large catchment with regular high rainfall and low evaporation – catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation – forest / vegetation in the flooded area decaying anaerobically releasing methane – access for workers and materials and distance to demand – value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • **Alternatives for supply:** rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • **Conservation strategies:** low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • **Evaluation:** reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane – conservation is cheap and quick but limited by behaviour – transfers create conflicts between regions – desalination has high energy / carbon cost – combination needed as demand grows • Justified conclusion	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **impermeable** → does not let water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

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Question	Marking guidance	Marks
11.2	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

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